RecoverLens

MOVE BETTER • RECOVER SMARTER

FUTURE HEALTH CHALLENGE · CHALLENGE 09

RecoverLens: Hack2skill upgrade plan

From an AI rehabilitation mirror to an adaptive recovery-adherence system: adapt the plan, detect a lapse, re-engage without blame and prove sustained change.

Challenge	Sustaining Healthy Behaviors
Lead innovator	Harry James Makoni
Build window	8 to 20 November 2026
Live app	recover-lens.vercel.app
Date	7 November 2026

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1. Summary

RecoverLens can enter Challenge 09, Sustaining Healthy Behaviors, as an adaptive recovery-adherence system. It is a weak entry today and a strong one only if the lapse-and-comeback loop is built before the deadline.

Positioning. Camera pose tracking is the sensor, and Gemini is the personaliser. The product is the loop: adapt the plan, detect a lapse, re-engage without blame and prove sustained change.

Why the fit is real. The behaviour is a rehabilitation exercise, which the challenge names as "recovery". Non-adherence to physiotherapy can reach 70% (Jack et al., 2010). RecoverLens already measures whether the exercise was done and done well, and it is built for language, disability and low-bandwidth access.

Why it is weak today. The app does not adapt the plan, does not re-engage a patient after missed sessions, and names no behaviour-change technique. It has no persistence data, because it has no real users yet.

What must be true to submit

- ☐ Registered on Hack2skill. A third-party report says registration closes on 10 November and prototypes are due on 20 November. I could not open the page, so confirm both dates there.
- ☐ The loop works end to end in the demo: check-in, adapted plan, session, lapse, barrier question, comeback.
- ☐ All cohort data is labelled as simulated everywhere it appears.
- ☐ A registered physiotherapist has reviewed the exercise library and the safety rules, or the entry makes no claim that it was reviewed.
- ☐ The privacy claims in the entry match what the app really sends to Gemini (section 5).

2. Fit against the five requirements

The upgrade turns three weak or missing requirements into working features and keeps the one RecoverLens already does well.

Requirement	RecoverLens today	What the upgrade adds
1. Adaptive personal plans	A physiotherapist sets exercises and target angles. The app compares each session with the patient's own baseline. The plan itself never changes.	Gemini builds the plan from a physio-approved library, scaled by completion and form score, with a plain-language "why this plan". A validator clamps it to safe limits.
2. Re-engagement after a lapse	Only the physiotherapist is alerted to missed sessions. The patient gets nothing.	A lapse engine fires after about 2 missed sessions and asks a one-tap barrier question: pain, time, motivation, forgot or no space. It then adapts length, difficulty, reminder time and channel. Progress is never reset to zero. Caregiver mode stays.
3. Evidence-based support with visible rationale	A voice coach, caregiver mode and WhatsApp sharing. No named technique and no stated limits.	Named techniques: goal-setting, self-monitoring, graded tasks and social support. A "Why am I seeing this?" panel shows the rationale and its limits. A per-user bandit learns which nudge works.
4. Autonomy and inclusion	**Strongest area.** English, chiShona and isiNdebele, WCAG 2.2 AA, voice guidance, offline use, no video upload, observation-only wording.	User control of goals, data and notifications. A language pack that can be extended. Low-bandwidth, SMS and WhatsApp channels. No guilt-based streaks. A written policy against sharing with employers or insurers.
5. Sustained outcome measures	Records completion, form, pain and range. No real users, so no persistence data.	A simulated cohort that reports persistence, recovery after a lapse, retention, functional change and results by subgroup, clearly labelled as simulated. A small real usability test if possible.

3. The core loop

The model personalises the plan, but two plain-code gates sit on either side of it: the red-flag rules before the model and the validator after it.

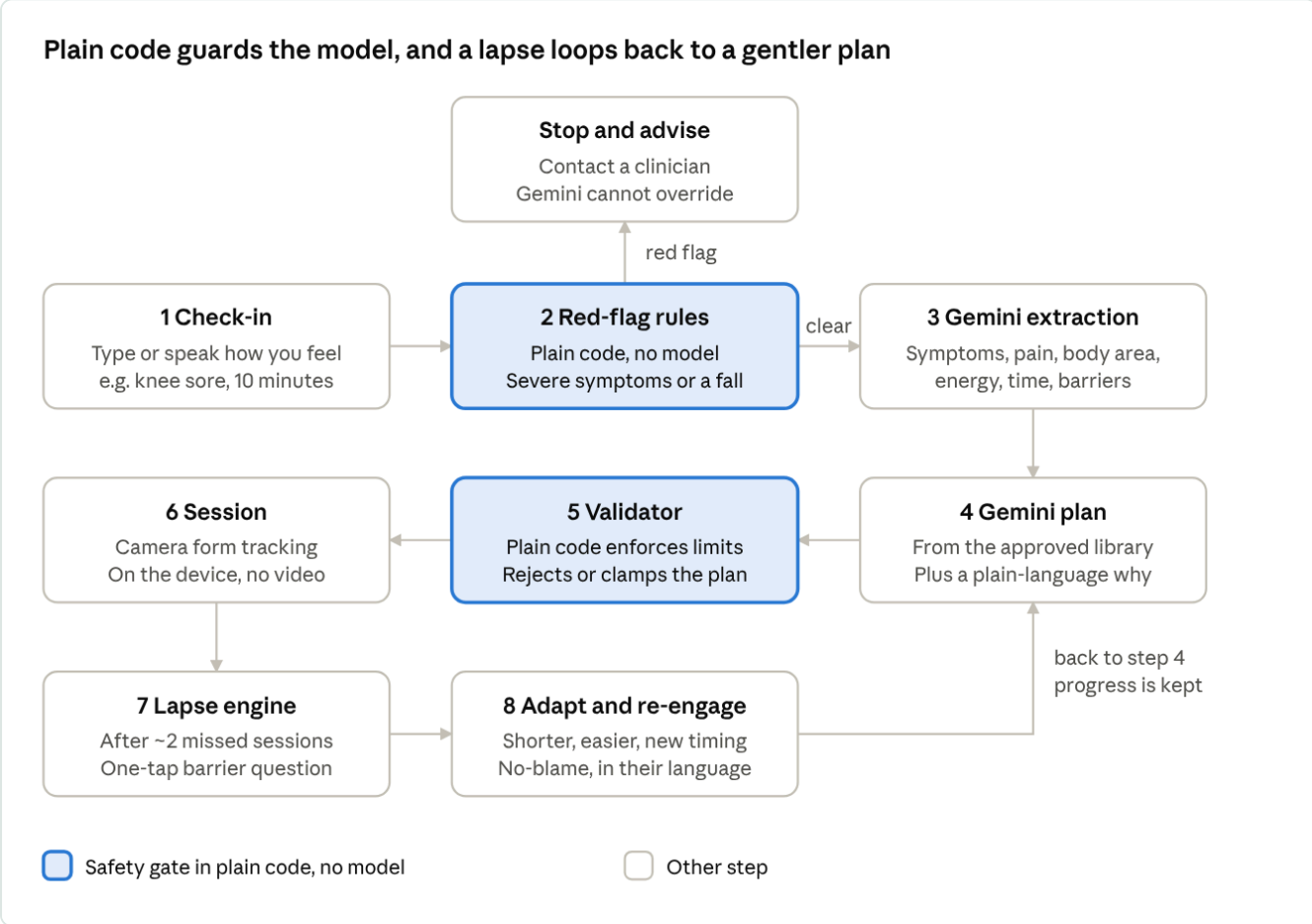


Figure 1. The adaptive recovery loop: two plain-code safety gates, one loop back.

A lapse returns the loop to step 4 with a gentler plan, and the user's progress is kept. Gemini writes the re-engagement message in the user's language, and plain code decides when it is sent and what the plan may contain.

4. What carries over and what is new

Most of the sensing, safety and language work already exists. The new build is the personalisation, the lapse loop and the proof.

Piece	Status	Notes
Camera pose tracking, rep counting, form score	Reuse	Runs on the device; no video stored
Exercise definitions as configuration objects	Reuse	The plan generator picks from these; it never invents exercises
Pain check-in before a session	Extend	Add free-text feelings, energy, time available and the red-flag rules
Personal baseline, fatigue signs, recovery projection	Reuse	Inputs to the plan and to the outcome metrics
Adherence (completed over prescribed in 14 days) and a clinician alert after 3 days without a session	Extend	The backend already computes this; the lapse engine moves the trigger to the patient and to about 2 missed sessions

Piece	Status	Notes
Offline queue, last-plan cache	Extend	Cache the latest plan so it works offline
English, chiShona, isiNdebele; WCAG 2.2 AA; caregiver mode	Reuse	The re-engagement messages need all three languages
Free-text extraction and plan generation with Gemini	New	Runs on a server endpoint so the API key never reaches the browser
Plan validator	New	Plain code; the only gate between the model and the patient
Lapse engine, barrier question, nudge bandit	New	The core of the entry
"Why am I seeing this?" panel	New	Names the technique and its limits
Consent screen and a "what is sent" view	New	Needed because text now leaves the device
Simulation and outcome dashboard	New	Labelled simulated everywhere

5. Safety, privacy and governance

Gemini can suggest, but plain code decides what a patient is allowed to do. Two parts of this design conflict with what RecoverLens says today, and both need a decision (section 10).

5.1 Safety

- **Red-flag rules run first, in plain code.** Severe pain, numbness, swelling, chest pain or a fall stops the session and advises the person to contact a clinician. Gemini cannot override this, and the wording stays observational, never a diagnosis.
- **Gemini works only inside a physio-approved library.** It selects and sequences approved exercises with sets, reps and intensity.
- **A validator enforces the limits.** It applies contraindications, maximum progression, session length and the allowed-exercise list, and it rejects or clamps anything outside them.
- **A physiotherapist can override any plan.** An audit log records the inputs, outputs and rationale of every plan.
- **It is recovery support, not diagnosis.**

5.2 Privacy

- **Video is never sent anywhere.** Pose tracking stays on the device.
- **Only de-identified text goes to Gemini.** Users see exactly what is sent before it leaves the device and give consent first. Free text can still contain a name by accident, so de-identification is best-effort: the check-in prefers one-tap chips, limits length and warns against names.
- **The API key stays on the server.** The browser calls our own endpoint, and the model provider sits behind a swappable interface.
- **A rules-only fallback mode** works with no model call, and the last plan is cached so the app works offline.
- **User control and a written policy.** Users control their goals, data and notifications. No data is shared with employers or insurers, and the entry says so in writing.

5.3 What changes in what we have already said

The ATU proposal and the privacy page say that only session numbers leave the device. Once check-in text goes to Gemini, that statement is no longer true. The privacy page, the in-app consent text and any ATU material must be updated in the same release.

6. Proof

With no real users yet, the entry proves the loop with a simulated cohort. A simulation shows that the system behaves as designed under stated assumptions. It does not show that the system works on people, and the entry must say that in plain words.

6.1 Design

- **Cohort:** 200 to 500 simulated users across five archetypes: busy worker, low motivation, caregiver-dependent, low literacy and disability.
- **Three arms on the same simulated people:** a static plan, generic reminders and RecoverLens.
- **Fixed before running:** the behaviour model, the definition of an active week (at least one completed session) and the metrics below. Nothing is tuned after seeing results.

Metric	Definition	Reported at
Persistence	Share of users with an active week	Weeks 4, 8 and 12
Return after a lapse	Share of lapsed users who complete a session within 7 days of the comeback message	Whole run
Retention	Share of users still active at the end	Week 12
Functional change	Change in range of movement against each user's baseline	Week 12
Equity	Gap between the best and worst archetype on persistence	Weeks 4, 8 and 12

6.2 Evidence that does not depend on the simulator

- **Unsafe plans blocked.** Real Gemini calls on simulated check-ins, counting how many plans the validator rejected or clamped, and why.
- **Physiotherapist appropriateness score** on a sample of generated plans. This needs a physiotherapist; without one, the entry reports no such score.
- **A small real usability test** with 10 to 20 people, if time allows.

6.3 Limits to state in the entry

- The nudge bandit learns from the simulator's response model, so its gains reflect that model. Report ranges under different assumptions, and show the cases where RecoverLens does not win.
- Every chart and number from the cohort carries the word "simulated".

7. Scalability and pilot path

The entry should separate what runs on 20 November from what is designed and what is roadmap, so a judge never mistakes one for another.

Capability	Status for the submission	Note
Works on low-end phones and offline	Built	Last plan cached; the rules-only mode needs no connection
Extensible language pack	Built for English, chiShona and isiNdebele	Each new language is a translation file
Standard progress export for clinicians	Designed	A FHIR-style JSON export; the CSV and print reports already exist
WhatsApp channel	Designed for the demo	A share link works today. Automatic messages need a WhatsApp Business account and approval.
SMS channel	Designed	Needs a provider and a server job
Health Connect and Apple Health import	Roadmap	The app is a web app, so it cannot read these directly; it needs a native wrapper

7.1 Pilot

- **Where:** physiotherapy clinics and workplace musculoskeletal programmes.
- **What it tests:** persistence over 12 weeks, return after a lapse and retention, with the same metrics as the simulation, so real results can be set beside the simulated ones.
- **Success metric, business model and regulation:** the entry states a success metric, the existing business model (practices and schools pay, patients with a physiotherapist use it free) and a note that RecoverLens is

positioned as wellness and recovery support rather than a medical device, to be confirmed with the regulator before any clinical claim.

8. Timeline

The build runs from 8 to 19 November, and the submission goes in on 20 November, or a day early if the demo video is ready.

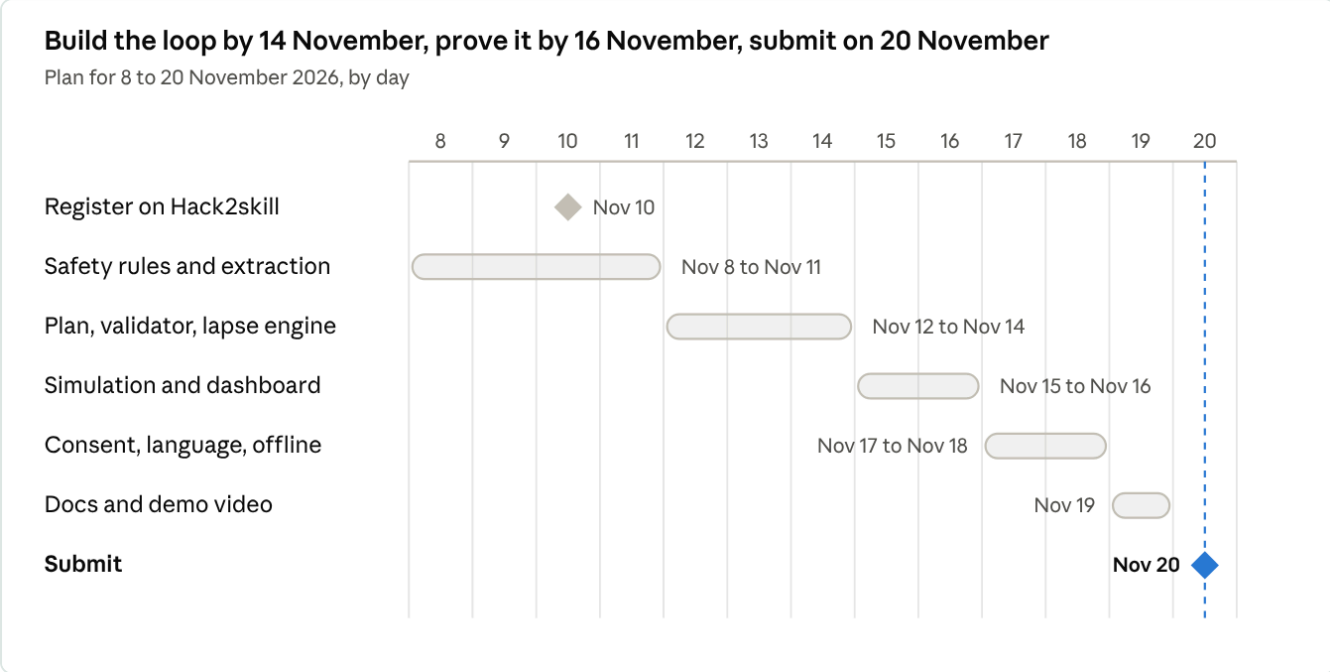


Figure 2. Build and submission plan, 8 to 20 November 2026.

The safety rules and extraction come first because everything else depends on them. The simulation starts only when the loop works end to end. Dates assume the Hack2skill deadlines in section 10 hold; confirm them before relying on this plan.

9. The 3-minute demo

The demo follows one simulated persona through a good week, a setback, a lapse and a return, and ends on the outcome dashboard.

Time	Scene	What the judge sees
0:00	The problem	One sentence: people stop exercising after a setback, and most programmes reset them to zero
0:20	A good week	The persona completes sessions with camera form tracking, on-device, no video stored
0:50	A setback	The persona types "knee sore, only 10 minutes". Gemini extracts the symptoms and builds a shorter, gentler plan from the approved library. The "Why am I seeing this?" panel names the technique and its limits.
1:30	The lapse	Two sessions are missed. A non-judgmental message arrives in the persona's language, with a one-tap barrier question.
1:55	The comeback	The persona answers "no time". The plan shortens and the reminder moves. The persona returns, and progress is kept, not reset.
2:15	The safety net	The persona types a red-flag symptom, and the session stops with advice to contact a clinician.
2:30	The proof	The outcome dashboard shows the three arms, simulated and labelled, with persistence at weeks 4, 8 and 12
2:50	The close	Privacy, languages and the pilot path in one line

10. Open decisions and risks

Six items need an answer before or during the build. The first two change the code, so they come first.

Item	Why it matters	Recommendation
Red-flag stop versus the project rules	CLAUDE.md rule 2 and SPEC §8.1 allow the pain check-in as the only interruption. A hard stop for chest pain, numbness or a fall goes beyond that.	Treat red flags as part of the pain check-in and extend §8.1 in the same commit. Show advice, not a diagnosis, and let the person correct a mistaken answer. This is your decision.
Privacy statements	The app today sends only session numbers. Check-in text going to Gemini breaks that claim in the privacy page and the ATU proposal.	Update both when the feature ships, and add the consent screen first.
A physiotherapist to review the library	"Physio-approved" is only true if a registered physiotherapist has reviewed the exercises, contraindications and progression limits.	Name the reviewer this week. If no one agrees in time, call the library "drawn from standard home-programme exercises" and make no review claim.
Dates and the registration form	I could not open the Hack2skill page. The 10 November registration and 20 November prototype dates come from a third-party report.	Read the page yourself today and send me the registration questions.
Gemini access and a persistent database	The model needs an API key on the server, and plan caching and the audit log need storage. Production has no database connected: it runs on temporary storage that resets.	Add the key as a server environment variable, never in chat. Connect Neon Postgres to the Vercel project before the demo.
Competing deadlines	ATU pitching runs to 10 November and the final is on 7–8 December.	Finish the ATU answers first. Do not let this plan delay them.

One optional item: the challenge comes from Abu Dhabi, and Arabic is not in the app. Adding an Arabic language pack would help the inclusion story, but it needs a native speaker to check it.

11. Cut list and the first build task

Anything that does not feed the loop or the proof waits until after 20 November.

11.1 Cut

- The five extra live exercises and any other new exercise types
- Visual polish beyond what the demo shows
- Health Connect and Apple Health import
- SMS delivery, and automatic WhatsApp messages beyond the share link
- Arabic, unless a native speaker is available

11.2 First task for Claude Code

Build the part that makes the loop safe before building the part that makes it smart:

1. **Pure TypeScript, no React:** the red-flag rules, the plan schema and the validator, with tests that cover contraindications, progression limits, session length and allowed exercises.
2. **The extraction schema:** JSON for symptoms, pain level, body area, energy, time and barriers.
3. **The Gemini prompts:** one for extraction and one for the plan with its "why this plan", both limited to the approved library.
4. **A server endpoint** that calls the model, passes the output through the validator and falls back to rules-only when the call fails.
5. **Checks:** `npm test`, `npm run build` and `pytest` all pass before each commit.

The registration text comes after the form questions are known.